



Faculty of Engineering and Computer Science
Department of Electrical and Computer Engineering

ETHICAL DIMENSIONS

ELEC/COEN 390 - BB

Electrical/Computer Engineering Product Design Project

Stakeholders

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**We certify that this submission is our original work and meets the
Faculty's Expectations of Originality**

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Automated Aquarium Monitoring Systems (AAMS Inc.)

AQUASENSE

Ethical Dimensions • Rev. 1.0.0

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1. Privacy Issues

Although water-quality readings collected by the app are not sensitive in themselves, our system will keep data that can reveal information about the user's home. Sensor timestamps and alert activity can reveal daily routines and whether a home is occupied. The device will also store the user's home Wi-Fi credentials. This sensitive information can cause extreme consequences if leaked. Strong system security (encrypted storage and transmission) will need to be in place to prevent data leaks; the app should be password-protected, and data must never be sold or shared with other parties without consent. The type of data being tracked will be clearly disclosed. We will collect only what is needed for tank monitoring (excluding location or personal details).

2. Potential for Misuse and Security

By automating water quality monitoring, AquaSense frees users from difficult, time-consuming manual testing and prevents tools from being disconnected without the user's knowledge. Real-time readings, instant critical-condition alerts, and easy-to-read analytics let aquarium owners stay on top of every tank and act before small changes threaten marine life. From data collection to alerts and graphical analysis, all in one place.

3. Alert Overuse

The app sends alerts to the user, notifying them of any issues with their aquariums. Notifications are sent based on any critical data being collected (values going above or below the thresholds, sensors offline, etc.). If notifications are sent too frequently, however, users could become desensitized and may ignore critical alerts, creating undesirable environments where emergencies go unnoticed. To fix this, Aquasense implements customizable notifications, where user's could select which types of notifications are critical.

4. Animal Protection

Users will rely on the app to safeguard living aquatic animals; therefore, sensor failures or inaccurate readings could lead to the animal's death. So, the app and sensors will require proper calibration. Sensors will be given a replacement date, and the app will remind users to calibrate and visually verify the sensors' conditions. If a sensor dies or the device goes offline, the app will raise a fault alert immediately rather than showing stale good data. The app will also state clearly what is and is not measured by the device and prompt users to log manual test-kit results, preventing a false sense of security.

5. Transparency and Accountability

Users will be made aware of how the data is collected and how accurate each reading is. It will be made clear that the product's intended purpose is to collect data and alert users; it is a decision aid, not a guarantee of fish safety. Alert thresholds (pH, temperature, etc.) are user-configurable, with system defaults that favor safety. As the app is strictly for data collection, any harm done to the fish due to poor conditions is the responsibility of the owner. If the app sends incorrect readings, the responsibility falls on Aquasense, and it is the company's obligation to portray accurate data. If incorrect data is shown and it leads to the animal getting sick or dying, the company bears the responsibility to financially compensate the client.

6. Data ownership and third parties

Cloud notifications rely on third-party infrastructure (e.g., Firebase). Users will be told what is stored, for how long, and that data will never be sold or used beyond the service. Users can delete their data and run the device offline if they prefer. Data is not owned by Aquasense; Aquasense is the means to collect the data. All data is owned by the user; therefore will not be shared without the user's consent.

7. Environmental

The product involves electronics, probes with consumable parts, and batteries. We will use replaceable probes and will document proper battery disposal.

8. Labor Sourcing

Electrical components such as sensors and microcontrollers will be purchased by third parties. To ensure all components are ethically sourced, parts will be purchased from manufacturers who abide by the international fair labor standards. Verifying manufacturers apply fair wages, safe working conditions, reasonable working hours, enforced health and safety laws. Audits of the manufacturing companies will be done to guarantee they comply with labor standards. As the company is still in its early stages, hardware assembly and packaging will be done internally until outsourcing becomes possible. Aquasense abides by the Canadian and Quebec labor laws, as the company is based in Montreal, Quebec.